

Predicting Unit Commitment in Short-Term Hydropower Scheduling

TDT4173 Modern Machine Learning in Practice — Course Project

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Abstract

We predict, for each of the 14 generating units of the Tokke–Vinje hydropower system and each hour of a one-week scheduling horizon, the probability that the unit is committed (on) in the solution of the SHOP optimiser. Our final model is a blend of four LightGBM models trained on a long-format table with one row per (case, unit, hour) and 180–207 engineered features. The decisive features encode the plant economics: the hourly price relative to the water value of the reservoir feeding the unit, and the length of the contiguous period during which that ratio stays above one, a proxy for start-up and shut-down costs. With a time-blocked, purged cross-validation the blend reaches a micro ROC-AUC of 0.9844 out of fold and 0.99045 on the public Kaggle leaderboard. We also report the ideas that did *not* work (sequence neural networks, stacking across units and cases, pseudo-labelling, using future reservoir volumes) and why.

1 Problem and evaluation

Short-term hydropower scheduling decides, hour by hour, how much each unit produces. The binary on/off *unit commitment* (UC) variables make the underlying mixed-integer programme expensive to solve; a good prediction of them can be used to fix or relax binaries and speed up the optimisation.

Each *case* is a 168-hour week starting at midnight UTC; a new case starts every day, so consecutive cases overlap by six days. The training set has 2922 cases (2015-01-01 to 2022-12-31) and the test set 725 cases (2023-01-01 to 2024-12-25). For every case we must output $14 \times 168 = 2352$ probabilities. The score is the *micro-averaged* ROC-AUC: all $725 \times 2352 \approx 1.7$ million predictions are pooled into a single ranking. Two consequences guided the work:

- predictions must be comparable *across units and weeks*, not only within one unit’s week;
- knowing which units are on (or off) for an entire week already gives a large share of the score.

The public leaderboard covers 2023 and the private one 2024, so the public score is only an estimate.

2 Data and exploratory analysis

Inputs. Hourly day-ahead price (EUR/MWh) and inflow to 17 reservoirs and one river; daily storage volume and water value (EUR/MWh) of the 17 reservoirs; hourly minimum-volume (3 reservoirs) and minimum-flow (6 rivers) constraints; and the SHOP topology file with reservoir capacities, generator limits, turbine curves and the connections between objects. We turned the raw tables into per-case arrays (`src/data.py`): price 168, inflow 18×168 , volume and water value 17×8 (days 0..7, day 7 being the end of the horizon), constraints, and labels 14×168 . There are no missing values.

Topology. From the connection list we derived, for each plant, the reservoir(s) it draws from and the reservoir it releases into (Table 1). This mapping is the backbone of the feature engineering.

Table 1: Plants, units and their reservoirs (from `Tokke_Vinje_topology.yaml`). $q_{\max}$ is the largest turbine discharge on the efficiency curves.

Plant	Units	Upstream reservoir(s)	Releases into	$P_{\max}$ [MW/unit]	$q_{\max}$ [m ³ /s/unit]
Tokke	4	Vinjevatn	Bandak	107.5	32
Vinje	3	Vaamarvatn	Vinjevatn	100	55
Songa	1	Songavatn, Bitdalsvatn	Totak	120	52
Kjela	1	Foersvatn	Hyljelihyl	60	40
Lio	1	Byrtevatn	river	43	14
Byrte	1	Botnedalsvatn	Byrtevatn	20	8
Hogga	1	Bandak	–	17	168.6
Vesle Kjela	1	Kjelavatn	Foersvatn	8.5	20
Haukeli	1	Vatjern	Vinjevatn	4.9	2.1

Label structure. About half of the (case, unit) pairs are constant for the whole week (Figure 1); in mixed weeks a unit switches only 2–6 times, i.e. decisions come in long blocks. Base rates differ strongly (Vinje G3 is on 26% of the hours, Lio 75%).

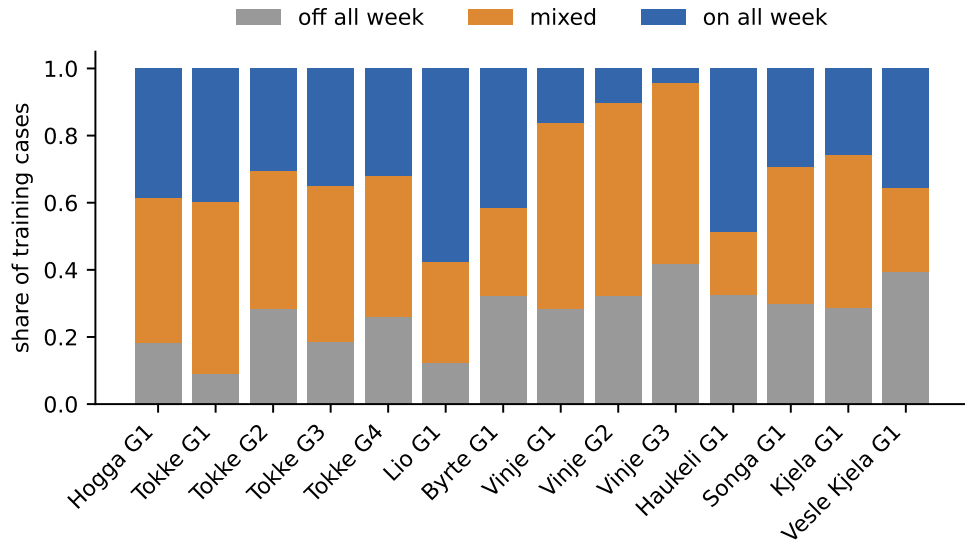


Figure 1: Weekly behaviour per unit on the training cases: share of weeks where the unit is off all week, on all week, or switches.

Label noise. Hour $t + 24$ of case d and hour t of case $d + 1$ refer to the same wall-clock hour, yet they agree only 88% of the time. The optimum depends on the horizon end (end-of-week water values), so the same hour is scheduled differently from one case to the next. Any model that ignores the case context is bounded by this.

Simple baselines. On the training set a per-unit constant (the unit’s base rate) gives 0.617 AUC, the within-week price rank 0.589, and price minus the water-value difference 0.539. The relationship is clearly non-linear and unit-specific.

Economic intuition. Figure 2 (left) shows the typical pattern: a Tokke unit runs when the price exceeds the water value of its upstream reservoir and stops during short, deep price dips. The right panel shows the hard case: with an almost flat price the decision is driven by water balance (Tokke releases into Bandak, so Hogga must eventually run) and the model misses it.

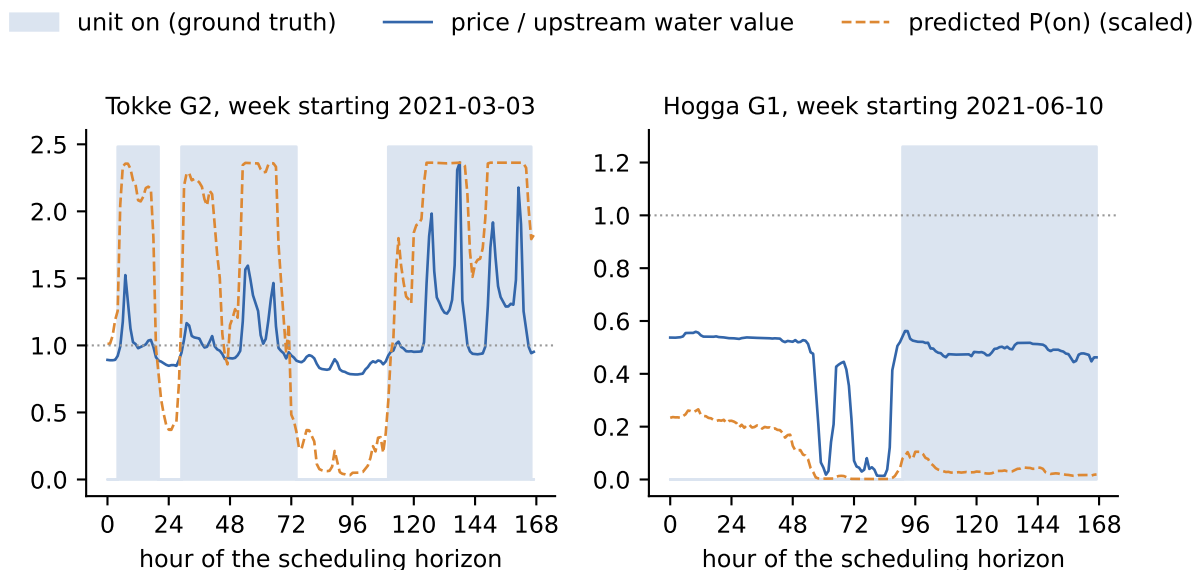


Figure 2: Two training weeks. Shaded: ground-truth commitment. Solid: price divided by the end-of-horizon water value of the upstream reservoir (dotted line at 1). Dashed: out-of-fold prediction of the final blend.

3 Validation strategy

Random K-fold would leak: a case and its neighbours share six of their seven days. We used four *time-blocked* folds of two years each (2015–16, 2017–18, 2019–20, 2021–22) and removed from training every case starting within 7 days of the validation block (purge). All numbers in this report are out-of-fold (OOF) micro ROC-AUC on the 2922 training cases unless stated otherwise. For quick experiments we trained on a 20% row subsample with a higher learning rate and evaluated only folds 2–3 (2019–22); repeating such a run with another seed moved the scores by about ± 0.0005 , which sets the threshold for calling an idea an improvement.

The CV turned out to be a reliable guide: the gain from `gbm_v2` to the blend was $+0.0016$ both out of fold and on the public leaderboard (Figure 3).

4 Feature engineering

The model works on a long table with one row per (case, unit, hour): $3\,647 \times 14 \times 168 \approx 8.6$ million rows (`src/features.py`). Features are grouped as follows (the version in which each group was introduced is given in brackets).

Price shape (v1). Raw price, z-score and ratio to the weekly mean, rank within the week and within the day, deviation from the daily mean, centred rolling means (3–25 h), differences to the previous/next hour, weekly quantiles; later (v2) mean of the next and previous 24 h.

Calendar (v1). Hour of horizon, hour of day, day of week, month and day-of-year (sin/cos).

Reservoir state (v1–v2). For all 17 reservoirs: initial filling ratio, end-of-horizon and initial water value (absolute and relative to the mean price); mean inflow per reservoir; constraint series.

Unit static (v1). Unit id (categorical), $P_{\min}$, $P_{\max}$, start cost, generator index within the plant, number of generators in the plant.

Plant economics (v1–v3). With $w^\uparrow$ the end-of-horizon water value of the upstream reservoir(s) and $w^\downarrow$ that of the downstream one: the price minus, and divided by, the difference $w^\uparrow - w^\downarrow$; **price over $w^\uparrow$** (`p_over_upwv`), the share of hours whose ratio exceeds $\{0.8, 0.9, 1, 1.1, 1.25\}$, and **rank_m_need_up** = price rank minus the rank threshold implied by that share (positive when the hour is among the hours worth producing in).

Run lengths (v3). Start-up and shut-down costs make it unprofitable to react to short spikes or dips. For the boolean series “ratio > 1” we compute the signed length of the run containing hour t (`seg_up_len`), the position inside it and the hours left; the same for “price > water-value difference”.

Water balance (v4/v5). Filling of the upstream and downstream reservoir if the unit does not release (initial volume plus cumulative natural inflow, optionally plus the maximal discharge of the plants feeding it), hours until full, hours to empty the reservoir at full discharge, room left downstream.

Cross-plant (v4 only). Every plant’s price/water-value ratio and run length at hour t , visible to every unit, to let e.g. Hogga see what Tokke is likely to do.

Future volumes were deliberately excluded. The volume table also contains the storage on days 1..7 of each horizon, which is available for the test period. Adding the weekly volume change lowered the 2021–22 fold from 0.9695 to 0.9653. The daily release implied by these volumes correlates weakly ($|r| \leq 0.24$) with the daily on-hours, so the volumes describe the historical operation of the system, not the SHOP solution we predict, and they act as noise.

5 Models

Gradient boosting (final). LightGBM [1] with binary log-loss, 255 leaves, at least 200 rows per leaf, feature fraction 0.5, bagging 0.7, $L_2 = 10$, learning rate 0.05, early stopping on the validation fold (150 rounds), 30–40% row subsampling to fit in 30 GB of RAM. The test model is refit on all training cases with $1.1\times$ the mean best iteration. A single model trains in 15–25 minutes on 8 CPU cores. We also used an Azure ML cluster (8 vCPU, student subscription) to run one of the full trainings in parallel.

Ensemble. Four GBMs that differ in feature set (v2, v3, v4, v5; Table 2) were averaged in logit space with weights 0.4/0.1/0.3/0.2 found by a grid search on the OOF predictions; equal weights give practically the same score (0.98431).

Neural networks (not used in the final submission). (i) A transformer over the 168 hours of a case with static context and a 14×168 output head, from raw inputs: 0.914 on 2021–22. (ii) A dilated temporal convolutional network over the GBM feature table, one sequence per (case, unit), receptive field above 168 h: 0.957 on 2021–22, with a training loss far below the validation loss. Both overfit the 2922 weeks. In a blend with the GBM they add at most +0.0008, and without a GPU (none was available in our quota) each fold took about 20 minutes, which ruled out further tuning.

6 Results

Table 2: Full four-fold results. Scores are micro ROC-AUC; Kaggle = public leaderboard (2023).

Model	#feat.	2015–16	2017–18	2019–20	2021–22	OOF	Kaggle
GBM v1 (price + reservoirs)	136	0.9813	0.9799	0.9546	0.9695	0.9716	–
GBM v2 (+ plant economics)	157	0.9861	0.9840	0.9745	0.9850	0.9826	0.98881
GBM v3 (+ run lengths)	172	0.9863	0.9836	0.9750	0.9863	0.9833	–
GBM v4 (+ water bal., cross-plant)	207	0.9874	0.9842	0.9702	0.9875	0.9830	–
GBM v5 (+ water balance)	180	0.9871	0.9843	0.9745	0.9860	0.9835	–
Blend v2 + v4		0.9873	0.9846	0.9757	0.9871	0.9842	0.99045
Blend v2 + v3 + v4 + v5		0.9873	0.9847	0.9763	0.9870	0.9844	0.99043

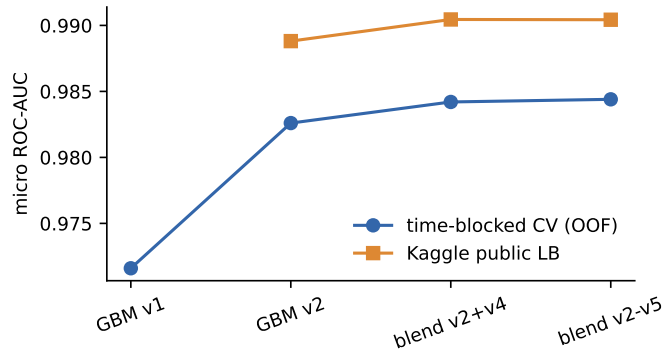


Figure 3: Out-of-fold CV and public leaderboard. Improvements in CV carried over to the leaderboard one-to-one.

Ablations and negative results. Table 3 lists every experiment we evaluated, including those that did not help. Most ideas that should help in principle (more context, more coupling between units, post-processing) did not beat the noise level of ± 0.0005 .

Error analysis. Figure 4 breaks the OOF score down. The weakest units are Haukeli (4.9 MW, tiny head reservoir Vatjern whose storage is not even recorded), Vinje G3 (the third unit, started only when G1 and G2 are not enough) and Hogga (driven by the water Tokke releases into Bandak). All three depend on what *other* units do, which the model only sees indirectly. 2020, a year with exceptionally low prices and high inflow, is clearly the hardest. Within mixed weeks the model ranks hours only slightly better than the price rank alone, so the remaining errors are mostly about where blocks start and end.

Model interpretation. Figure 5 shows the split-gain importance. The model is dominated by `p_over_upwv` and the run length `seg_up_len`, followed by the unit identity and the “need rank”. Physically this means: a unit runs when power is worth more than the water it uses, and only if that holds long enough to pay for starting it. Cross-plant ratios of the two big plants (Tokke, Vinje) and the water value of the small Vinjevatn reservoir, where both plants meet, are the next most informative signals.

7 Discussion and limitations

- **Independence assumption.** Every (unit, hour) is predicted on its own. Units are coupled through the watercourse, and our attempts to model this (stacking, cross-plant features,

Table 3: Experiments on folds 2019–20 / 2021–22 (quick setting) or on the full OOF. ▲ adopted, ● neutral, ▼ rejected.

Idea	Result	
Future weekly volume change	2021–22: 0.9695 $\rightarrow$ 0.9653	▼
Plant economics (price / water value, need rank)	2019–20: 0.9546 $\rightarrow$ 0.9736; 2021–22: 0.9695 $\rightarrow$ 0.9845	▲
Run lengths above threshold	0.9736 / 0.9845 $\rightarrow$ 0.9744 / 0.9856	▲
Cross-plant features	0.9744 / 0.9856 $\rightarrow$ 0.9707 / 0.9864	● (diversity)
Water balance	0.9744 / 0.9856 $\rightarrow$ 0.9741 / 0.9853	● (diversity)
Lags/leads and windowed max/min of the ratio	$\rightarrow$ 0.9738 / 0.9848	▼
Stronger regularisation (63 leaves, 2000 rows/leaf)	$\rightarrow$ 0.9748 / 0.9848	●
Stacking with neighbouring cases’ predictions (same hour)	every fold below its base model	▼
Stacking with all units’ predictions	OOF 0.9718 vs. 0.9826 base	▼
Temporal smoothing of the logits	best 0.98436 vs. 0.98437	▼
Per-unit logit recalibration	0.9837 vs. 0.9844	▼
Pseudo-labelling the unlabeled period	matches its teacher; +0.00005 in blend	●
Transformer / TCN sequence models	0.914 / 0.957 alone; $\leq$ +0.0008 in blend	▼
Ensembling GBMs with different feature sets	OOF 0.9835 (best single) $\rightarrow$ 0.9844; Kaggle +0.0016	▲

joint neural networks) either overfit or did not transfer across years. A joint sequence model trained on a GPU is the most promising direction we could not explore properly.

- **Distribution shift.** Year-to-year scores range from 0.961 (2020) to 0.989; the private leaderboard (2024) may therefore differ from the public one.
- **Irreducible variability.** Overlapping cases agree only 88% on the same hour. Part of the remaining error is due to the optimiser’s sensitivity to the horizon end and to near-equivalent MIP solutions.

8 Conclusion

Feature engineering grounded in the physics and economics of the watercourse was what moved the score: expressing the price relative to the water value of the reservoir that feeds each unit (0.9716 $\rightarrow$ 0.9826) and adding the length of the profitable period (0.9833). A blend of gradient-boosting models with different feature sets gave the final 0.9844 OOF / 0.99045 public AUC. A purged, time-blocked validation was essential: it predicted the leaderboard gains exactly and let us reject many attractive but useless ideas.

Reproducibility. All code is in `src/`; see Section A.

References

- [1] G. Ke et al., “LightGBM: A highly efficient gradient boosting decision tree,” in *Advances in Neural Information Processing Systems 30*, 2017.
- [2] SINTEF Energy Research, *SHOP — Short-term Hydro Operation Planning, documentation*, <https://docs.shop.sintef.energy>.

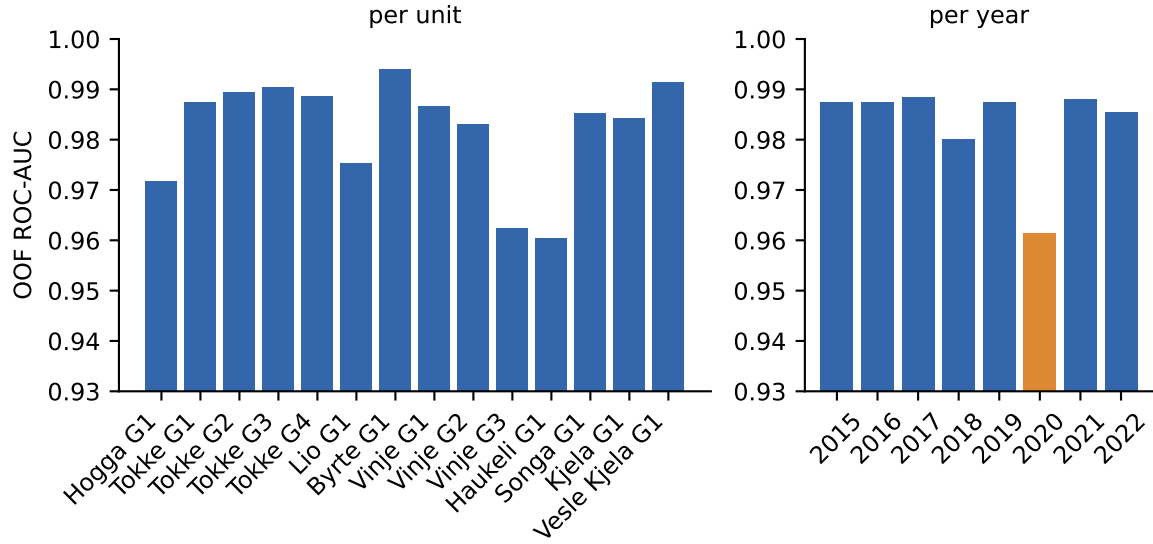


Figure 4: Out-of-fold AUC of the final blend per unit and per year.

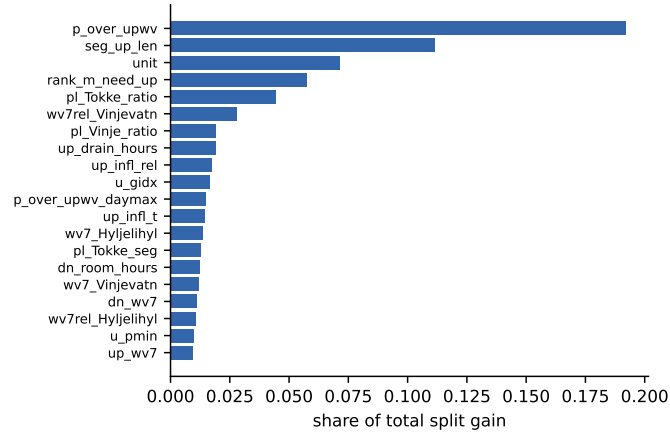


Figure 5: Top-20 features by LightGBM split gain (v4 feature set, model trained on 2015–20).

A Reproducing the results

```

uv venv --python 3.12 .venv
uv pip install --python .venv pandas numpy scikit-learn lightgbm \
    pyyaml pyarrow matplotlib
.venv/bin/python src/data.py # build cache/cases.npz
# gbm_v2 was trained before later feature groups were added (commit b3d90c8)
.venv/bin/python src/train_gbm.py --name gbm_v3 --sub 0.3 \
    --no-future-vol --no-cross --full
.venv/bin/python src/train_gbm.py --name gbm_v4 --sub 0.3 \
    --no-future-vol --full --seed 4
.venv/bin/python src/train_gbm.py --name gbm_v5 --sub 0.3 \
    --no-future-vol --no-cross --full --seed 5
.venv/bin/python src/blend.py gbm_v2 gbm_v3 gbm_v4 gbm_v5 # weights on OOF
.venv/bin/python src/submit.py gbm_v2:0.4 gbm_v3:0.1 gbm_v4:0.3 \
    gbm_v5:0.2 --out submissions/blend4.csv
.venv/bin/python report/make_figures.py # figures of this report

```